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Cooling history and age of magnetization of a deep intrusion: A new 1.7 Ga key pole and Svecofennian-post Svecofennian APWP for Baltica

Sten-Åke Elming[a,*], Paul Layer[b], Ulf Söderlund[c]

[a] Division of Earth Sciences, Luleå University of Technology, SE-97187 Luleå, Sweden
[b] Geophysical Institute, University of Alaska Fairbanks, AK, USA
[c] Department of Geology, Lund University, Sölvegatan 12, SE-22362 Lund, Sweden

ABSTRACT

A paleomagnetic and chronogical study has been performed on the Turinge gabbro-diabase formation and on a cross cutting basic dyke in central Sweden and on the Joulovaara gabbro intrusion in northern Sweden in the Fennoscandian Shield. U-Pb age of baddeleyite and $^{40}Ar/^{39}Ar$ ages of hornblende and biotite reveal a cooling history of the deep gabbro-diabase intrusion in Turinge. The cooling is suggested to have taken place in two stages, one related to the time of intrusion in temperature down to ca 500 °C with a cooling rate up to 46–59°/Ma and another at a lower rate of ca 2.9 °C/Ma, which is suggested to be related with uplift. From this cooling history it can be concluded that the magnetization age of the diabase, ca 1695–1700 Ma, is close to the crystallization age and the $^{40}Ar/^{39}Ar$ age of hornblende. Applying a similar cooling history for the other studied deep intrusion, the ca 1800 Ma gabbro of Joulovaara gabbro, it is estimated that the magnetization age of the gabbro should be close to that of the U-Pb age of the formation, although the pole of the Joulovaara gabbro is less reliable.

The cooling history presented here for the Turinge gabbro-diabase has implications for estimations of magnetization ages also for other deep intrusions.

The new pole (Plat. = 51.6°, Plon. = 220.2°; A95 = 4.8°) of the Turinge gabbro-diabase passes most of the reliability criteria and is considered a new key pole for Fennoscandia.

The Basic dyke that cuts the Turinge gabbro-diabase was here dated at ca 1200 Ma (whole rock, $^{40}Ar/^{39}Ar$) and the virtual geomagnetic pole calculated from its primary magnetization falls into the expected trend of APWP for Baltica.

The new Turinge key pole prolong the time of overlapping poles for Fennoscandia, indicating only small movements of the shield between ca. 1870 to 1700 Ma.

1. Introduction

The presence of 2100–1800 Ma orogenic events in most of the continents (e.g. Zhao et al., 2004) has led many researchers to propose the assembly of a Mesoproterozoic supercontinent, named Nuna, Colombia or Hudsonland (e.g. Hoffman, 1997; Meert, 2002; Pesonen et al., 2003, Zhang et al., 2012) that exsisted already during the Paleoproterozoic. This supercontinent included Laurentia, Fennoscandia/Baltica and possibly India (Zhang et al., 2012) and cratons of Australia (e.g. Karlstrom et al. (2001)). Paleomagnetic analyses suggest that Nuna was formed at ca 1650 Ma ago (Pisarevsky et al., 2014) and possibly as early as at ca. 1750 Ma (e.g. Salminen et al., 2016).

According to Bogdanova et al (2008) and Elming et al (2001, 2010) Baltica was not assembled until ca. 1800 Ga ago when Sarmatia (Ukrainian Shield) and Volgo-Uralia amalgamated with Fennoscandia. Sarmatia subsequently rotated into its present position in Baltica (Elming et al., 2010), i.e. at the time of formation of the supercontinent Nuna.

Paleomagnetism in combination with radiometric datings is the only quantitative method for reconstructing past movements of the continents and assembly of supercontinents. For reliable reconstructions at the time of assembly of the Nuna supercontinent, the paleomagnetic poles must pass certain reliability criteria (Van der Voo, 1990) to be defined as reliable key poles. For Fennoscandia/Baltica there are only a few key-poles for the time interval 1900–1600 Ga of the suggested assembly of Nuna. Paleomagnetic poles in the 1880–1786 Ma time period are overlapping indicating only small movements of Fennoscandia. Those poles are to a large extent based on paleomagnetic data from

* Corresponding author.
E-mail address: sten-ake.elming@ltu.se (S.-Å. Elming).

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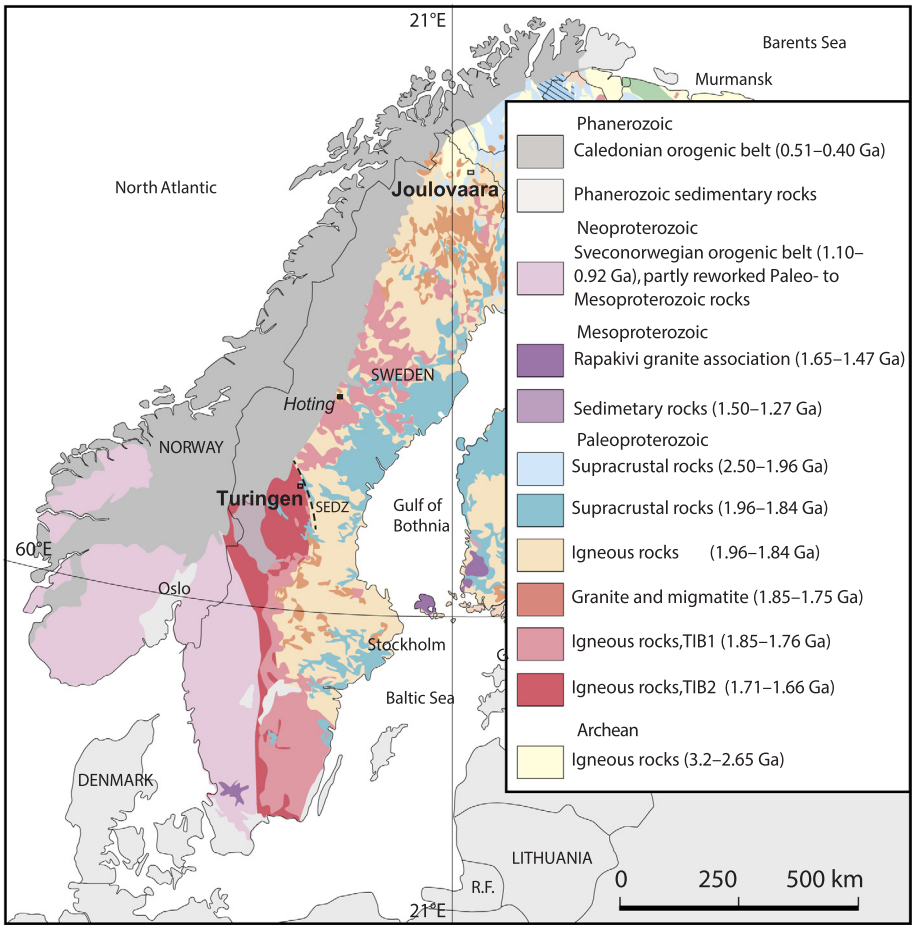


Fig. 1. The geology of the Proterozoic basement of the Fennoscandian Shield in Sweden (modified from Bergman et al., 2012) and Finland and the location of the studied Joulovaara gabbro and the Turinge diabase. The hatched line SEDZ denotes the Storsjön-Edsbyn Deformation Zone and TIB1 stands for the 1.76–1.81 Ga Transscandinavian Igneous Belt group 1 rocks.

gabbroic rocks that intruded at great depth and for which the ages of magnetization in relation to their crystallization ages are poorly constrained. Here we make paleomagnetic studies of basic rocks from central and northern Sweden, one of which coupled with radiometric age determinations, in an attempt to determine the cooling history of a deep intrusion relative to its age of magnetization.

With this new study we expect to fill the time gap of reliable paleomagnetic poles in the interval 1800–1650 Ma for Baltica.

2. Geological background

The Fennoscandian Shield is built up of a number of tectonomagmatic blocks representing a geochronological zonation and accretionary growth, the oldest parts being found in the northeast and the youngest in the southwest (Gaál and Gorbatschev, 1987).

In Sweden, Archean basement rocks are found only in the very northern part of Sweden (Fig. 1). In the Paleoproterozoic, 2300 to 2100 Ma (e.g. Skiöld, 1987), supracrustal belts were formed during the rifting stages of the Archean craton and new continental crust was formed during the 1930 to 1870 Ma Svecofennian orogeny (Skiöld, 1987, Vaasjoki and Sakko, 1988). This was followed by the intrusion of vast amounts of granitoids that were emplaced 1800–1770 Ma ago.

The Joulovaare gabbro (Figs. 1 and 2) was intruded into the belt of ca 2300–2050 Ma old supracrustal rocks and ca 1.9 Ga old granite/granodiorite east of the Archean core in northernmost Sweden, separating Archean gneisses in the west from the younger Svecofennian rocks in the east. A syenite-monzonite intrusion, inter fingering with the gabbro, has been dated at ca. 1800 Ma (based on a concordant U-Pb age on zircon; Hellström and Bergman, pers. comm., 2017).

The Turinge gabbro-diabase is located in the boundary between the Transscandinavian Igneous Belt (TIB) and the Central Svecofennian province (Fig. 1). The 1860–1600 Ma TIB generally follows the western and southwestern margins of the 1920–1860 Ma crust formed during the Svecofennian orogenic belts (Fig. 1) (Högdahl et al., 2004). The boundary between TIB and the Central Svecofennian Province is marked by a tectonic belt, namely the Storsjö-Baltic Sea Zone, which in the area of Turingen coinsides with the Storsjön-Edsbyn Deformation Zone (SEDZ; Högdahl and Sjöström, 2001).

The Turinge gabbro-diabase has a tholeiitic composition, it is medium to fine grained and dominated by plagioclase with a labradore composition (55–60% An). Clinopyroxene and orthopyroxene are generally common, while titanomagnetite and biotite are subordinate, but with grains of biotite up to 0.01 m in diameter (Lundqvist et al., 1990). The gabbro-diabase is located 10–15 km east of the Rätan granite, which forms a part of TIB (Fig. 1) and the granite shows no signs of regional metamorphism (Lundqvist et al., 1990). The gabbro-diabase is intruded into predominantly metamorphosed and deformed Svecofennian supracrustal rocks (Fig. 3). The eastern parts of the gabbro-diabase are close to the SEDZ, however there are no signs of deformation in the rock.

The gabbro-diabase is a ca 20 km long and 4 km wide formation (Fig. 3) and gravity data indicate a depth extension of at least 3 km (Delin and Aaro, 1997). The gabbro-diabase is cut by granitic dykes, which are interpreted to be related to the Rätan granite (dated by U-Pb zircon at 1698 +25/−20 and 1679 +70/46 Ma, Wilson et al., 1985) but the source could also be assimilated melts from the melted host rocks at the time of intrusion of the gabbro-diabase indicating that the formation was emplaced at great depth. This may also be indicated by the quartz- potassium feldspar hybrid rocks located in the upper part of the gabbro-diabase (Lundqvist et al., 1990). A pegmatite dike cutting the south-western part of the diabase has been dated at 1650 ± 15 Ma ($^{40}Ar/^{39}Ar$-biotite age; A.M. Kähr in Lundqvist et al., 1990). Since large

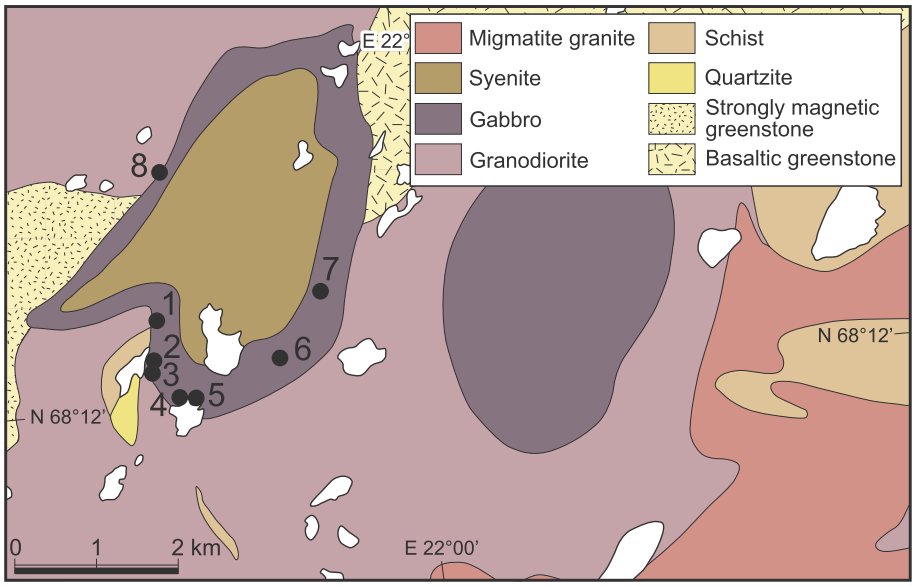


Fig. 2. The ring shaped Joulovaara gabbro and the paleomagnetic sampling sites (•). The gabbro intruded into 2.3–2.05 Ga greenstone, basaltic greenstone and 1.91–1.88 Ga granite/granodiorite (Bergman et al., 2012). Note the Syenite/monzonite in the center of the intrusion, which has been dated at ca 1800 Ma (Hellström and Bergman, personal communication).

inclusions of late orogenic rocks (ca 1800 Ma) are found in the gabbro-diabase, this and the age of pegmatite intrusion bracket the age of the diabase between 1800 and 1650 Ma. Basic dykes cutting the the Turinge gabbro-diabase are rare, but there is one 1.5 m wide dyke, which has a mineral composition similar to that of post Jotnian basic intrusions in the Fennoscandian Shield, the Central Scandinavian Dolerite Group (e.g. Gorbatschev et al., 1979; Söderlund et al., 2006). The dyke, here named the Basic dyke, trends NW (330°E) and dips 70°W.

3. Sampling and laboratory technique

The samples were collected both as block samples and drill cores using a portable drill. Orientation of the samples was done with sun and magnetic compasses. In the Joulovaara gabbro 31 samples were collected from 8 sites (Fig. 2). Altogether 74 samples were collected in 6 sites of the Turinge gabbro-diabase and 17 from the Basic dyke cutting the gabbro-diabase (Fig. 3). The Basic dyke forms a good opportunity for a baked contact test. From the gabbro-diabase, samples were taken at increasing distances up to 6.0 m from the contact to the basic dyke. This was done in order to define areas baked and areas unaffected by the dyke intrusion.

The cores were cut into at least two specimens and the specimens were demagnetized by thermal and alternating field (AF) techniques. The thermal demagnetizations were done in steps of 50 °C up to 500 °C and 10 °C up to 600 °C, while the AF demagnetizations were done in steps of 5 to 10 mT up to 100 mT. The measurements of remanent

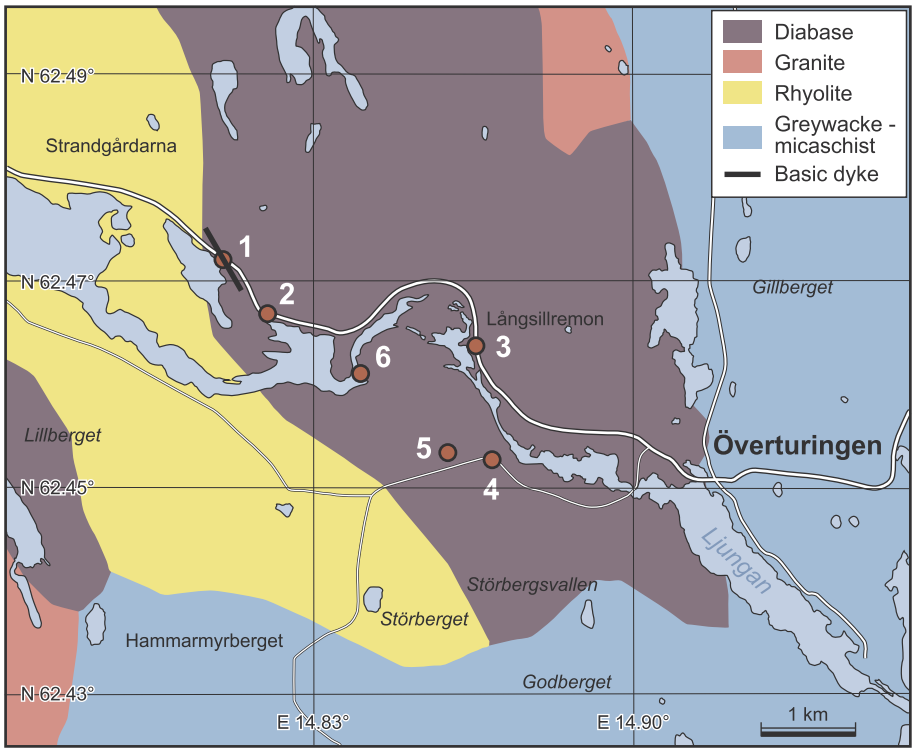


Fig. 3. The geology of the area of the Turinge gabbro-diabase and the sampling site for the paleomagnetic and geochronological studies. The gabbro-diabase intruded into ca 1.86 Ga rhyolite and granite/granodiorite and 1.91–1.87 Ga metamorphic sandstone, Greywacke (Bergman et al., 2012).

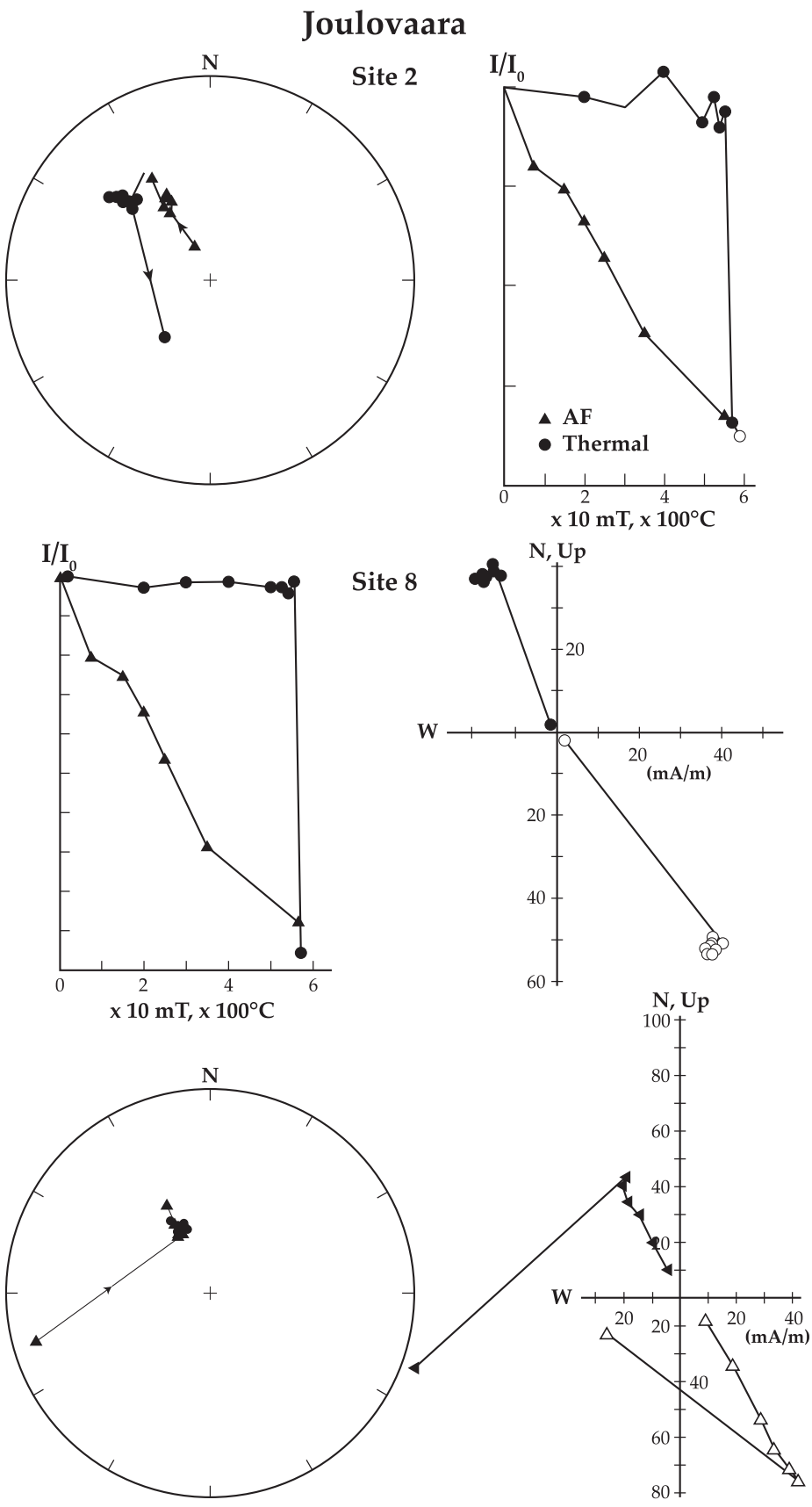


Fig. 4. Examples of remanence behavior from the gabbro at Joulovaara. The sample was first demagnetized by alternating field, followed by thermal demagnetization. In the vector plot the solid (open) symbols denote the end-point of the vector plotted in the horizontal (vertical) plane.

Table 1
Paleomagnetic results from the gabbro-diabase and basic dyke in Turinge and from the gabbro in Joulovaara.

Rock	Site	Glat/Glon N°/E°	N/n	Decl. (°)	Incl. (°)	α95 (°)	K	Plat. (N°)	plon (E°)	A95	Age (Ma)
Turinge											
Gabbro-diabase	1	62.475/14.813	1/17	350.3	37.6	5.3	45.6	48.1	208.4	4.8	
	2	62.470/14.823	1 /7	336.0	46.7	3.7	270.8	52.1	230.6	3.9	
	3	62.466/14.869	1/13	341.8	39.4	7.0	35.8	48.0	220.4	6.5	1703 ± 6; 1687 ± 13 (^{40}Ar/^{39}Ar, Hbl)
	4	62.451/14.869	1/6	338.8	45.3	7.9	73.5	51.7	226.2	8.0	
	5	62.452/14.856	1/6	346.1	46.7	7.0	92.0	54.3	216.2	7.2	
	6	62.457/14.838	1/7	342.9	48.2	4.6	176.7	54.9	221.4	4.8	1700 ± 4 (U-Pb, Bd)
Mean gabbro-diabase			6*/56	342.8	44.1	4.8	199.3	51.6	220.2	4.8	
Basic dyke	1	62.475/14.813	1/17	16.1	−50.0	3.4	111.5	4.2	1.0	3.7	1205 ± 4; 1194 ± 5 (^{40}Ar/^{39}Ar, WR)
Baked gabbro-diabase	1	62.475/14.813	1/15	12.1	−48.5	5.8	44.7	2.5	4.3	6.2	
Mean Basic dyke + baked gabbro-diabase			1/32*	15.3	−49.2	3.1	67.6	3.4	1.6	3.3	
Joulovaara											
Gabbro	2	68.203/21.918	1/4	328.2	47.2	8.5	117.6	46.0	243.8	8.8	Ca. 1800 (U-Pb, zr)
	3	68.202/21.918	1/3	338.6	41.4	12.0	106.7	43.7	229.4	11.4	
	5	68.200/21.925	1/3	330.2	15.9	18.6	44.8	26.7	235.4	13.8	
	6	68.202/21.957	1/2	319.5	32.5			33.4	249.8		
	8	68.222/21.923	1/5	335.0	48.0	11.7	43.5	48.1	235.5	12.4	
Mean gabbro			5*/17	330.0	37.3	13.9	31.4	39.1	239.0	12.5	

Notes: Glat./Glon. is the geographical latitude and longitude of the sampling sites. N is the number of sites and n is the number of samples. *denotes the level of mean calculation. Decl., Incl., is the mean declination and inclination, respectively. α95 is the radius of the cone of confidence at a 95% probability level and κ is the precision parameter. Plat., Plon., denotes the latitude and longitude of the paleomagnetic pole position and A95 id the radius of the 95% confidence circle of the poles. In the age data Bd, Bi and Zr stand for baddeleyite, biotite and zircon, respectively. The Joulovaara gabbro was dated by Hellström and Bergman (Pers. comm., 2017), while the ages of the Turinge rocks are from the present study.

Table 2
$^{40}Ar/^{39}Ar$ whole rock and mineral ages of the basic dyke and diabase in Turinge.

Age determinations					
$^{40}Ar/^{39}Ar$ and U-Pb					
Site	Formation	Mineral	Integrated age (Ma)	Plateau age (Ma)	Plateau information
$^{40}Ar/^{39}Ar$					
1	Basic dyke	Whole rock#1	1253 ± 4	1205 ± 4	Well defined plateau 57% ^{39}Ar release, MSWD = 2.8
		Whole rock#2	1221 ± 4	1194 ± 5	Well defined plateau 59% ^{39}Ar release, MSWD = 2.3
2	Turinge gabbro-diabase	Biotite	1626 ± 11	1627 ± 11	Well defined plateau 99% ^{39}Ar release, MSWD = 0.7
3	Turinge gabbro-diabase	Hornblende#1	1701 ± 6	1703 ± 6	Well defined plateau 94% ^{39}Ar release, MSWD = 1.1
		Hornblende#2	1684 ± 5	1687 ± 13	Not as good plateau as run #1 87% ^{39}Ar release, MSWD = 5.0
		Biotite #1	1613 ± 4	1622 ± 5	Plateau age, 62% ^{39}Ar release, MSWD = 2.3
		Biotite #2	1639 ± 4	1645 ± 4	Plateau age, 73% ^{39}Ar release, MSWD = 2.1
U-Pb					
5		Baddeleyite	1700 ± 4		Weigthed mean, MSWD = 2.4 (n = 4)

magnetization were performed using a SQUID (2G-DC) magnetometer. The remanence vectors were defined by least square fit of vectors in 3-D space (Torsvik et al., 1996) and by principle component analyses (Kirschvink, 1980).

$^{40}Ar/^{39}Ar$ age determinations have been carried out on the Turinge gabbro-diabase in the sites 2 and 3 and on the Basic dyke of site 1 (Fig. 3). Paleomagnetic specimens were submitted for $^{40}Ar/^{39}Ar$ analysis at the Geochronology Laboratory at the University of Alaska, Fairbanks. Hornblende and biotite grains were extracted for mineral dating and for the whole rock the specimens were crushed, washed and sieved to obtain small chips for dating. The monitor mineral hb3GR (Zartman, 1964) with an age of 1071 Ma was used to monitor neutron flux (and calculate irradiation parameter, J). The sample and standards were wrapped in aluminum foil and loaded into aluminum cans of 2.5 cm diameter and 6 cm height. The samples were irradiated in position 5c of the uranium enriched research reactor of McMaster University in Hamilton, Ontario, Canada for 20 megawatt-hours. Upon their return from the reactor, the samples and monitors were loaded into 2 mm diameter holes in a copper tray that was then loaded in an ultra-high vacuum extraction line. The monitors were fused, and samples heated, using a 6-watt argon-ion laser following the technique described in York et al. (1981), Layer et al. (1987) and Layer (2000). Three splits of the hornblende were run. Argon purification was achieved using a liquid nitrogen cold trap and a SAEZ Zr-Al getter at 400 °C. The samples were analysed in a VG-3600 mass spectrometer at the Geophysical Institute, University of Alaska, Fairbanks. The argon isotopes measured were corrected for system blank and mass discrimination, as well as calcium, potassium and chlorine interference reactions following procedures outlined in McDougall and Harrison (1999). System blanks generally were 2×10^{-16} mol ^{40}Ar and 2×10^{-18} mol ^{36}Ar, which are 10 to 50 times smaller than fraction volumes. Mass discrimination was monitored by running both calibrated air shots and zero-age glass sample. Duplicate runs were done on the samples for consistency. A summary of the $^{40}Ar/^{39}Ar$ results is given in Table 2, with all ages quoted to ± 1 sigma level and calculated using the constants of Steiger and Jäger (1977). The integrated age is the age given by the total gas measured and is equivalent to a potassium-argon (K-Ar) age. The spectrum provides a plateau age if three or more consecutive gas fractions represents at least 50% of the total gas release and are within two standard deviations of each other (Mean Square Weighted Deviation (MSWD) < ca. 2.5).

U-Pb datings of baddeleyite were done on a block sample from site 6 of the Turinge gabbro-diabase. Baddeleyite was separated using Wilfley water-shaker table following the procedures of Söderlund and Johansson (2002), followed by removal of magnetic minerals using a pencil magnet. Baddeleyite grains were of excellent quality and up to ca 100 μm in longest dimension. Five of the largest grains were hand-picked under binocular microscope and transferred to Teflon dissolution capsules. The grains were washed in 2–3 N nitric acid, washed repeatedly in a 10:1 HF:HNO_3 solution at 210 °C over 3 days. After dried down on a hot plate the samples were redissolved in 10 drops 6.2 N HCl together with a small of H_3PO_4. The samples were loaded together with silica gel onto an outgassed single Re spectrometer at The Museum of Natural History in Stockholm. Pb and U isotopic data were collected in the 1210–1240 filament temperature range. The SEM detectors mass discrimination correction factor was constant at 0.1% per atomic mass unit. The total procedural blank was estimated at 0.5 pg Pb and 0.05 pg U. The intensities of ^{208}Pb, ^{207}Pb, ^{206}Pb and ^{205}Pb were measured in static mode using Faraday cup collectors with the ^{204}Pb measured simultaneously in a ion counter equipped with RPQ filter. The translation of counts per second to mV was achieved by successively measurements of a 5–8 mV ion beam in one of the Faraday detectors and the SEM. This was performed between all sample measurements as a check on amplification stability. Initial common Pb was corrected for using the model composition of Stacey and Kramers (1975). Decay constants are those of Jaffey et al. (1971). All age errors quoted in the text and Table 3 and in error ellipses in the Concordia diagram (Fig. 10), are given at the 95% confidence level. Plotting and age calculations are from Ludwig (2003).

4. Results

4.1. Paleomagnetism and baked contact test

In the Joulovaara gabbro the natural remanent magnetization (NRM) is carried by magnetite as indicated by thermal demagnetizations with unblocking temperatures in a narrow range at ca 580 °C (Fig. 4). The coercivity of remanence varies with medium destructive fields (MDF) between 5 and 50 mT. The characteristic remanent magnetization (ChRM) is isolated in samples from 5 of the 8 sites. The directions are defined with Mean Angular Deviations (MAD) less than 8° and the mean of sites (Table 1; Fig. 5) forms a formation mean at Declination = 330.0° and Inclination = 37.3° (α_{95} = 13.9°).

The Turinge gabbro-diabase, sampled in 6 sites, carries a natural remanent magnetization (NRM) with varying coercivities (Hc) expressed by MDFs in the range 5–70 mT. The ChRM is isolated in high Hc

Table 3
U-Pb baddeleyite age data from the Turinge diabase.

Analysis no. (number of grains)	U/Th	Pbc/Pbtot[1]	$^{206}Pb/^{204}Pb$	$^{207}Pb/^{235}U$	± 2 s % err	$^{206}Pb/^{238}U$	± 2 s % err	$^{207}Pb/^{235}U$	$^{206}Pb/^{238}U$	$^{207}Pb/^{206}Pb$	± 2 s	Concordance
	raw[2]			[corr][3]				[age, Ma]				
Bd-1 (1 grain)	82.0	0.040	1664.2	4.2916	0.53	0.29928	0.51	1691.7	1687.7	1696.7	3.5	0.995
Bd-2 (2 grains)	226.8	0.012	5476.7	4.3043	0.25	0.29936	0.19	1694.2	1688.1	1701.7	3.0	0.992
Bd-3 (1 grain)	213.8	0.010	6331.0	4.3172	0.22	0.30024	0.15	1696.6	1692.5	1701.7	2.9	0.995
Bd-4 (1 grain)	272.2	0.041	1549.2	4.2955	0.42	0.29948	0.27	1692.5	1688.7	1697.2	5.9	0.995

[1] Pbc = common Pb; Pbtot = total Pb (radiogenic + blank + initial).
[2] Measured ratio, corrected for fractionation and spike.
[3] Isotopic ratios corrected for fractionation (0.1% per amu for Pb), spike contribution, blank (0.5 pg Pb and 0.05 pg U), and initial common Pb. Initial common Pb corrected with isotopic compositions from the model of Stacey and Kramers (1975) at the age of the sample.

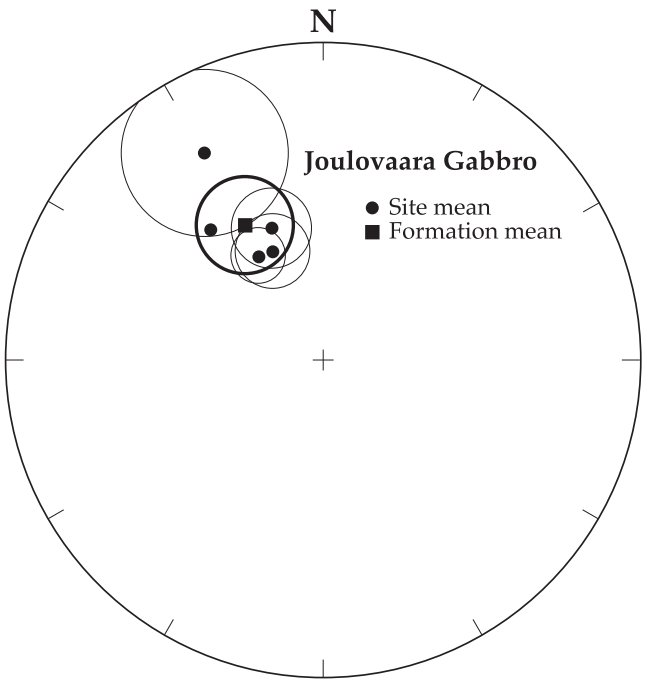


Fig. 5. Site mean and formation mean directions for the Joulovaara gabbro shown with the α_{95} confidence cones. The solid symbols represent downward projections of the mean directions and the full drawn cone of confidence refers to the formation mean.

and unblocking temperature (T_{ub}) ranges (> 500 °C, Fig. 6), although, the T_{ub} ranges are often wide (530–570 °C). From the thermal demagnetizations magnetite-titanomagnetite is indicated to be the principle remanence carrier (Fig. 6). The directions are well defined with MAD in the range 3.0°–7.2° and a mean direction of the 6 site means for the gabbro-diabase is calculated at Declination = 342.8° and Inclination = 44.1° (α_{95} = 3.4°; Table 1, Fig. 7).

Thermal demagnetizations indicate that the remanent magnetization of the Basic dyke is carried by magnetite (Fig. 6). Well defined ChRM directions are defined from both AF and thermal demagnetizations with MAD < 5.6° and a mean for the dyke including AF and thermal data is calculated at declination = 16.1° and inclination = −50.0° (α_{95} = 3.4°; Table 1, Fig. 7).

Baked contact tests for the Basic dyke are performed on samples collected along three profiles from the gabbro-diabase. A Basic dyke direction is obtained by thermal demagnetizations up to a distance of 1.55–2.30 m from the contact to the dyke. At distances from 1.55–2.3 m to 2.54 m (the hybrid zone) the remanent magnetization is often characterized by a large portion of low T_{ub} (0–400 °C) remanence (see Fig. 6D) with a direction of Present Earth Field (PEF). At higher temperatures (500–550 °C) a remanence vector is isolated with a direction (decl. = 0.7° and incl. = −43.6°) similar to that of the Basic dyke (Table 1, Fig. 7) and at temperatures higher than 550 °C a direction (decl. = 346.5° and incl. = 34.4°; Fig. 6D) similar to the remanence direction of the gabbro-diabase (Table 1, Fig. 7) can be isolated. At distances larger than 2.54 m the ChRMs carries a pure gabbro-diabase direction. The ChRM mean direction of the baked gabbro is calculated from 15 samples at Dec. = 12.1 and Inc. = −48.5° (α_{95} = 5.8°; Table 1, Fig. 7) and the combined basic dyke and baked host rock directions form a mean at Decl. = 15.3 and Incl. = −49.2 (α_{95} = 3.1°; Table 1).

4.2. Geochronology

In one sample from the Turinge gabbro-diabase at site 3 both hornblende and biotite were dated by the $^{40}Ar/^{39}Ar$ method. A single hornblende run gives a well-defined plateau with an age of 1703 ± 6 Ma. A second run resulted in a slightly lower age (1687 ± 13 Ma), but the plateau is more scattered, possibly reflecting alteration (Fig. 8; Table 2). Also the dating of biotite reveal plateau ages in the two runs, both ages lower than those of hornblende. The biotite plateau ages of the two runs are different from each other (1622 ± 5 Ma and 1645 ± 4 Ma; Fig. 8), possibly related to differences in grain size, the larger grain is older than the smaller one. A biotite grain from the gabbro-diabase at site 2 was also dated, with a well-defined plateau age of 1627 ± 11 Ma. This results in a mean $^{40}Ar/^{39}Ar$ age of biotite of 1631 Ma.

The basic dyke intruding the Turinge gabbro-diabase is dated from whole rock separates (Fig. 9; Table 2). The two runs reveal well-defined plateau ages of 1205 ± 4 Ma and 1194 ± 5 Ma.

The U-Pb analyses of four fractions of baddeleyite, of which three were single-grain analyses, plot less than 1% discordant (Fig. 10; Table 3) forming a tight cluster immediately below the Concordia curve. We favour a weigthed mean of $^{207}Pb/^{206}Pb$ dates as the best estimate for the crystallization age of the Turinge gabbro-diabase, which corresponds to 1700 ± 4 Ma. U/Th is exceptionally high for these baddeleyite grains (ca. 80–270), showing baddeleyite's high affinity of U over Th.

5. Discussion

5.1. Cooling history and age of the magnetization

The closing temperature of U-Pb system in baddeleyite is ca 900 °C (Söderlund et al., 2004), while the closing temperature of the Hb in the $^{40}Ar/Ar^{39}$-system is in the range 580–490 °C depending on the cooling rate (from 500 to 5 °C/Ma; Harrison, 1981). This mean that the cooling from ca 900 °C to 580–490 °C was relatively fast, within the error bars of the age data, 1700 ± 4 (Bd-age), 1703 ± 6 Ma (Hb-Ar/Ar) and 1687 ± 13 Ma (Hb-$^{40}Ar/^{39}Ar$), i.e. 320–410 °C/ca 30 Ma and a minimum cooling rate of ca. 11–14°/Ma if the less reliable 1687 Ma Hb-age is considered. A cooling rate of ca 46–59°/Ma, i.e. 320–410 °C/ca

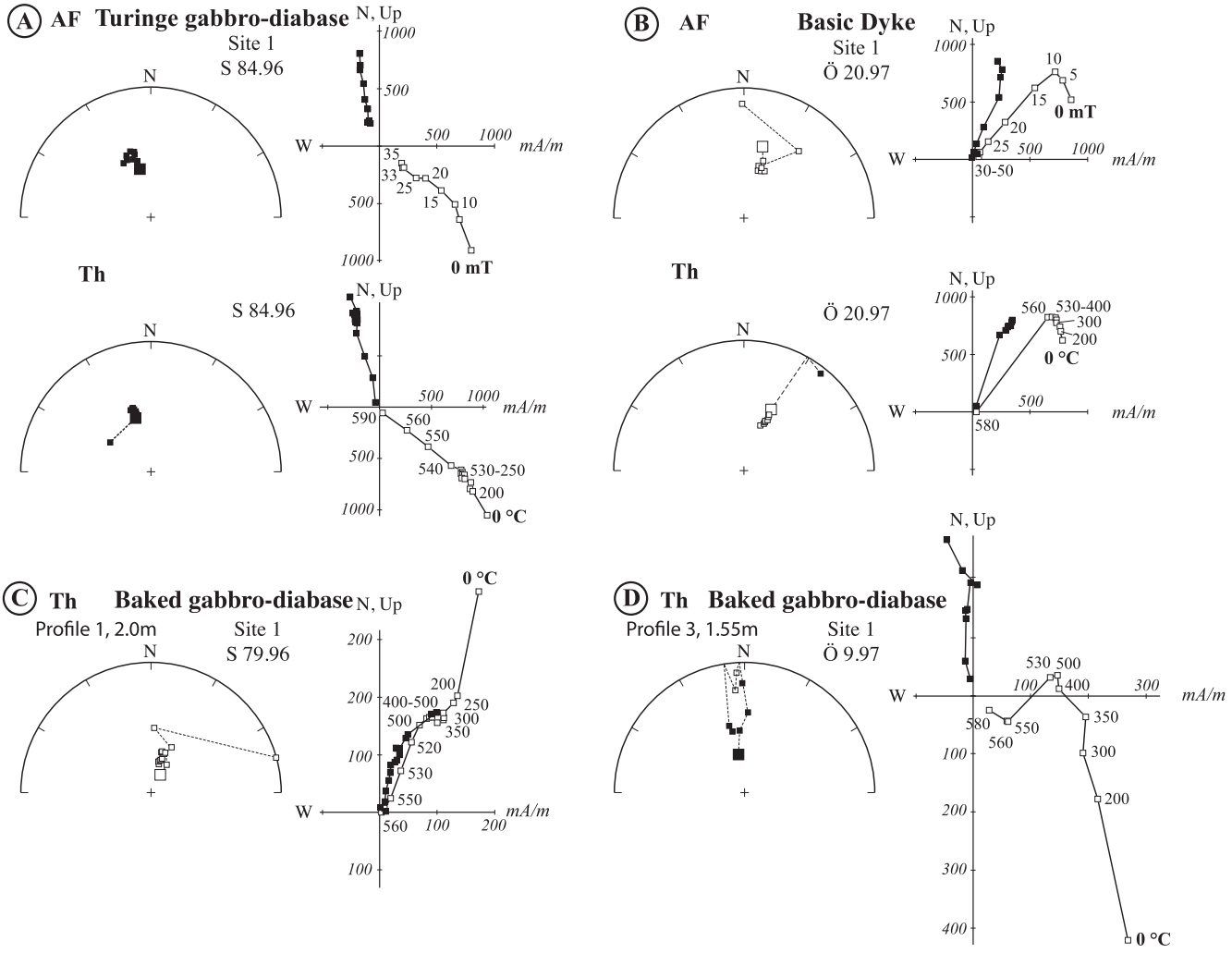


Fig. 6. Examples of remanence behavior during AF and thermal demagnetization for the Turinge gabbro-diabase, basic dyke and baked gabbro-diabase. The codes S 84.96 etc are sample names. In figures C and D the distances from the dyke along different profiles are given. In the stereographic plot the solid (open) symbols denote downward (upward) projections. For the vector plots see the conventions in Fig. 4.

7 Ma (1704–1697 Ma), is estimated if only the 1700 ± 4 (Bd-age) and the more reliable 1703 ± 6 Ma (Hb-age) is used for the calculation. This suggests a closing temperature for the $^{40}Ar/^{39}Ar$ of Hb in the range of 500 °C. For the Bi-Ar/Ar-system the closing temperature is in the range 310–280 °C for cooling rates of 10 to 1 °C/Ma (Harrison et al., (1985)). From the difference between the Hb- and Bi- $^{40}Ar/Ar^{39}$ ages of ca 70 Ma in the Turinge diabase the cooling rate is estimated low with closing temperatures of Hb of ca 500 °C and Bi of ca 300 °C, which will result in an estimated cooling rate of 2.9 °C/Ma. This can be compared with the cooling rate 3.5 °C/Ma estimated for a previously studied gabbro in the Hoting area (Fig. 1; Elming et al., 2009) and should be related with a slow uplift and cooling. A great depth of intrusion (in Hoting ca 10.4 km) for the Turinge diabase is also supported by up-melted host rock and granitic dykes cutting the gabbro-diabase, dykes which originate from melting of the host rock (Lundqvist et al., 1990).

The relatively fast cooling at the time of intrusion was thus followed by a slow cooling related to uplift (Fig. 11). Even if the cooling rate was relatively high during the blocking of the ChRM, the time of cooling is still considered long enough to average out secular variations in the magnetic field. The unblocking temperatures of the ChRM in the gabbro-diabase is in the range 530– 560 °C, i.e. close to the Cuire temperature of magnetite and similar to a natural blocking temperature for such cooling rate (Dodson and McClelland-Brown, 1980). This means that the magnetization age should be close to the baddeleyite (U-Pb) and hornblende ($^{40}Ar/Ar^{39}$) ages of ca 1700 Ma. This small time difference between the U-Pb age and age of magnetization is different from the ca 80 Ma that was estimated for the 1786 Ma Hoting gabbro (Elming et al., 2009), based on an assumed linear trend of cooling from the time of crystallization to the $^{40}Ar/^{39}Ar$ biotite age. There is no $^{39}Ar/^{40}Ar$ hornblende age for the Hoting gabbro. A similar age difference between the U-Pb baddeleyite and the $^{39}Ar/^{40}Ar$ biotite ages is also noted for the Turinge gabbro-diabase. Since both the Hoting gabbro and the the Turinge gabbro-diabase intruded at a great depth, their cooling histories are most probably similar. Based on the indicated relatively fast initial cooling at the time of intrusion of the Turinge gabbro-diabase, the magnetization age of the Hoting gabbro is now suggested to be close to 1780 Ma.

For the basic dyke cutting the diabase the $^{40}Ar/^{39}Ar$ whole rock ages are very consistent, however the first run (1205 Ma) may be more reliable including 94% Ar release. A quite fast cooling of the 1.5 m thick dyke through the narrow blocking temperatures (at ca 580 °C) of the magnetite carrying the ChRM may be expected and a ca 1200 Ma age of magnetization seems well motivated. The remagnetization of the host gabbro-diabase, where the rock carries only a Basic dyke direction, varies for the different profiles with distance from the Basic dyke contact between ca 1.55 to 2.3 m. This variation might be explained by differences in heat transfer, i.e. both through conduction and the effect of water circulation, although the latter is not expected be of great

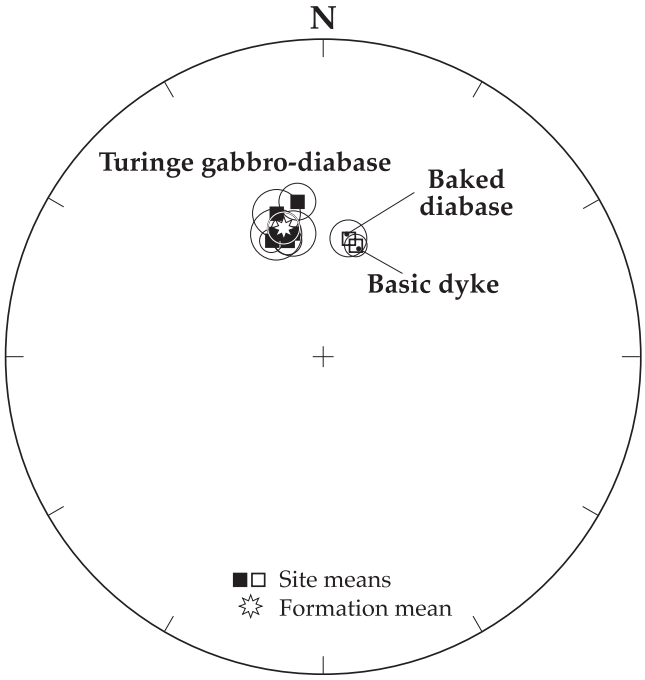


Fig. 7. Site mean directions for the Turinge gabbro-diabase, the basic dyke and the baked gabbro-diabase host rock. The solid (open) symbols denote downward (upward) projections.

significance in a crystalline rock. Another possible reason for the variation might be related to differences in domain structure. There is no data to support this but it could serve as a hypothesis to be tested in a future study. The distance to the beginning (ca 1.55–2.3 m) and to the end of the hybrid zone (ca 2.55 m) can be compared with what has been observed for a 1.35 m thick dyke cutting the Hoting gabbro. The hybrid zone there starts at ca 1.3 m and extends to at least 2.3 m from the contact to the dyke (Elming et al., 2009), i.e. similar to what is observed in the Turinge gabbro-diabase.

Magnetic susceptibility measurements were carried out on samples from the baked rock over the hybrid zone into the gabbro-diabase carrying pure host rock directions. These measurements do not reveal any significant changes in susceptibility, which may indicate that there was no chemical alteration of the baked gabbro-diabase and that the origin of the ChRM is thermal.

For the Joulovaara gabbro there is only one U-Pb age on zircon (Hellström and Bergman, personal communication 2017) to relate with the magnetization age. As for the Turinge diabase the time of magnetization age of the gabbro should be lower than but not so different from the U-Pb age 1800 Ma.

5.2. A post Svecofennian key pole for Fennoscandia and a loop in the APWP

The magnetization of the Turinge gabbro-diabase is reliably dated at ca 1700 Ma. The number of samples (56) and the precision of the mean direction (α_{95} = 4.8°; κ = 199) well meet the demand for a reliable paleomagnetic pole. The samples are adequately demagnetized and analysed. The diabase is cut by a basic dyke dated at 1200 Ma for which a positive baked contact test is demonstrated, which means that the Turinge gabbro-diabase has not been remagnetized since at least ca 1200 Ma ago. There is no resemblance of paleopoles younger than 1700 Ma and older than 1200 Ma and the subvertical dip of the Basic dyke cutting the gabbro-diabase suggest a structural control. Thereby 6 of the 7 criteria of Van der Voo (1990) are fulfilled, the presence of reversals is missing. The Turinge paleomagnetic pole (Plat. = 51.6°N, Plon. = 220.2°E; Tables 1 and 4) can be considered as a key paleomagnetic pole.

Poles of similar age have previously been reported from porphyries and granites of Dalarna county (Plat. = 82.0°, Plon. = 178°E; Bylund and Elming, 1992) and also for the neighboring Rätan granite (Plat. = 67°N, Plon. = 190°E; Mattsson and Elming, 2001). These poles, of not very high reliability, are located at higher latitudes and it is possible that a Present Earth Field (PEF) component has not been erased properly.

For the Joulovaara gabbro there is no field test to support a primary origin of magnetization. The gabbro is dated and with the cooling model from Turinge the age of magnetization should be only slightly lower than the U-Pb age, i.e. ca 1800 Ma. The samples have been adequately demagnetized and remanence vectors isolated, but the number of samples (17), from which the mean direction is calculated, is not enough to fulfill the criteria of Van der Voo (1990). With reference to the Turinge gabbro-diabase there is resemblance of younger paleopoles even if there is no U-Pb ages similar to that of the Turinge gabbro-diabase in the very northern part of Sweden. With reference to this, the pole (Plat. = 39.1°N, Plon. = 239.0°E; Tables 1 and 4) cannot be considered very reliable.

Paleomagnetic poles of ca 1870–1750 Ma to a large extent overlap (Fig. 12; Table 4). The new 1700 Ma Turinge paleomagnetic key pole (pole 12, Fig. 12) is located north of and with partly overlapping A_{95} confidence circles of the 1870–1750 group of poles but it is significantly different from the ca 1800 Ma Joulovaara pole (Fig. 12). Even if there may be a weak trend of a loop in the APWP key poles from the 1870 Ma Keuruu dyke pole (pole 2, Fig. 12, Table 4) through the 1786 Ma Hoting (pole 11) and 1780–1760 Ma Shoksha pole (pole7) to the more northern located Turinge pole (pole 12) the number of key-poles in this time interval is too few to truly define any loop. The big shift in pole positions comes from the ca 1700 Ma Turinge pole to the 1642–1621 Ma Wiborg Quartz porphyry dykes (Table 4), the 1633 Ma Sipoo and the 1578 Åland poles.

Some of the gabbro formations dated in the time interval ca 1900–1700 Ma seem to be overprinted, e.g. the pole from the Dundret gabbro (pole 10, Fig. 12; Elming, 1985). The gabbro has been dated at 1883 Ma and 1798 Ma, but the pole plots in between the 1633 Sipoo (pole 13) (Mertanen and Pesonen, 1995) and 1575 Åland (pole 15) (Salminen et al., 2016). This suggests that the gabbro in Dundret has been overprinted at ca 1600 Ma ago, which may be supported by U-Pb data from the adjacent Malmberget iron ore in which ages around 1600 Ma were traced (Romer, 1996; Storey et al., 2007).

The pole of the 1784–1769 Ma old Småland intrusions (Pisarevsky and Bylund, 2010) is considered to be a key pole. This pole (pole 6) is located close to the 1641–1621 pole of the Wiborg Quartz porphyry dykes (pole 14) rather than within the ca 1870–1700 Ma group of poles. A positive inverse baked contact test was suggested to indicate a primary remanent magnetization. However, the age of the cutting dyke is not known and if the age of the dyke is lower than ca 1550 Ma, the dyke may have intruded into already remagnetized Småland intrusions and the pole may thus be of a different age.

The pole calculated from the ca 1200 Ma basic dyke cutting the diabase is calculated from a ChRM of primary origin defined from a positive baked contact test. The pole (Plat. = −3.4°N, Plong = 181.6°E; Table 1) is a Virtual Geomagnetic Pole (VGP) since secular variations are not averaged out during the fast cooling of the dyke. The pole plots as might be expected between the Post Jotnian 1270–1246 Ma key-pole (Fig. 12; Elming and Pesonen, 2010) and the1095–995 Ma pole of the Laanila Dyke (Mertanen et al., 1996).

5.3. The accretion of Sarmatia/Volgo-Uralia with Fennoscandia

Paleomagnetic reconstructions are based on coeval poles from the different blocks or continents. Bogdanova et al. (2008) suggested a collision between Fennoscandia and Sarmatia/Volgo-Uralia at ca 1800 Ma. Paleomagnetic data from the Ukrainian Shield indicate that Sarmatia was not in its present position relative Fennoscandia at time

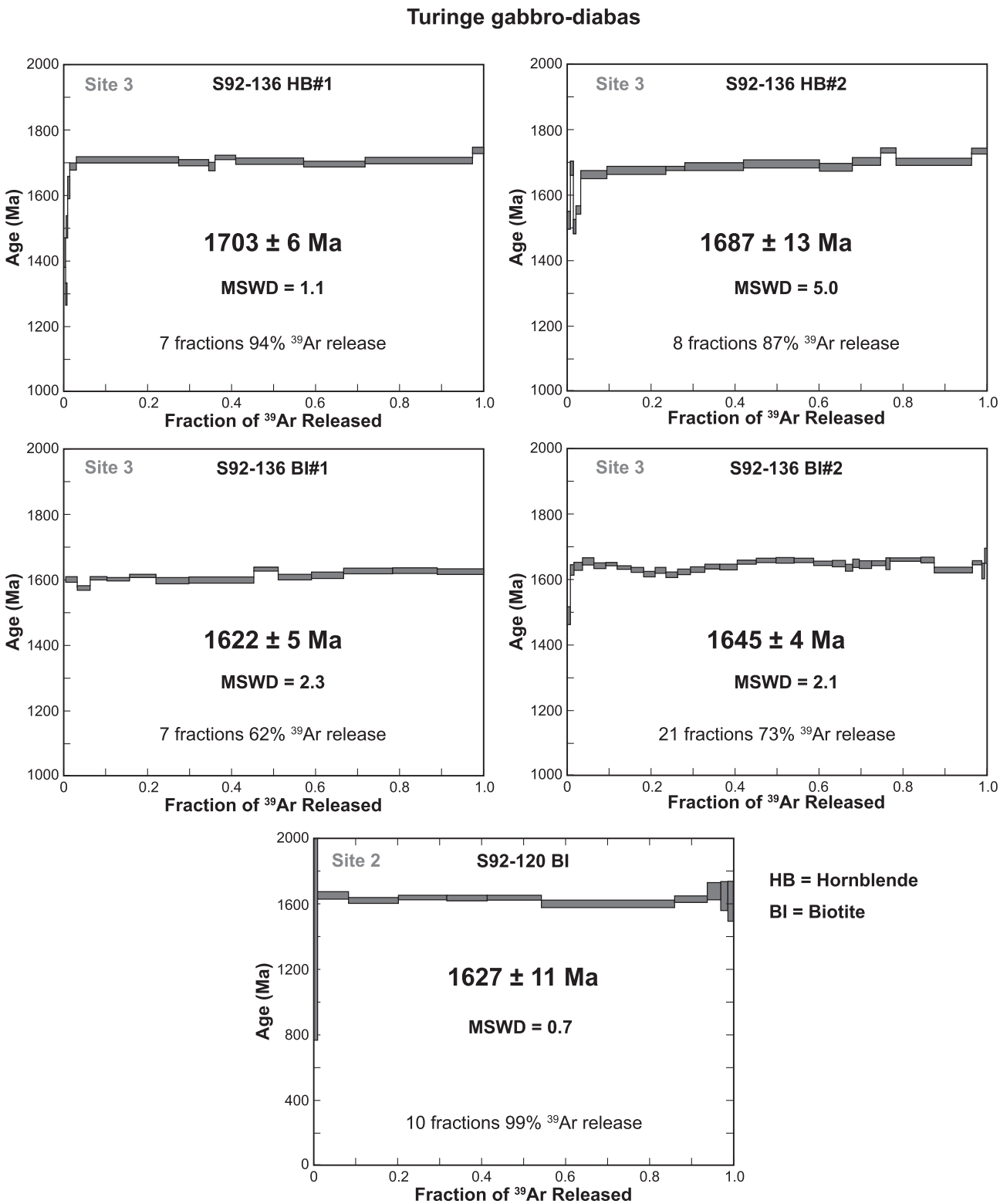


Fig. 8. $^{40}Ar/^{39}Ar$ age spectra from hornblende (HB) and biotite (BI) samples from the sites 2 and 3 of the Turinge gabbro-diabase. The samples are collected from the paleomagnetic drilling cores. See text for analytical procedures.

1720 Ma and earlier (Elming et al., 2010). For the tectonic reconstruction coeval ca 1770 Ma poles of Fennoscandia and Ukraine are used. Accepting that Sarmatia docked with Fennoscandia at ca 1800 Ma paleomagnetic reconstructions show that Sarmatia after collision rotated clockwise (Euler pole: 48°N, 42.2°E, 43° rotation angle; Elming et al., 2010) into its present position. With the new 1700 Ma pole for Fennoscandia a comparison with an almost coeval pole 1720 Ma for the Ukraine (Sarmatia) may suggest that the clockwise rotation continued another ca 50 Ma after ca 1770 Ma. There is no field test to prove a primary origin of the magnetization of the 1720 Ma Ukrainian pole. If the age of the 1720 Ma pole still is correct, a continued rotation of Sarmatia from its once rotated position, demands a new euler pole

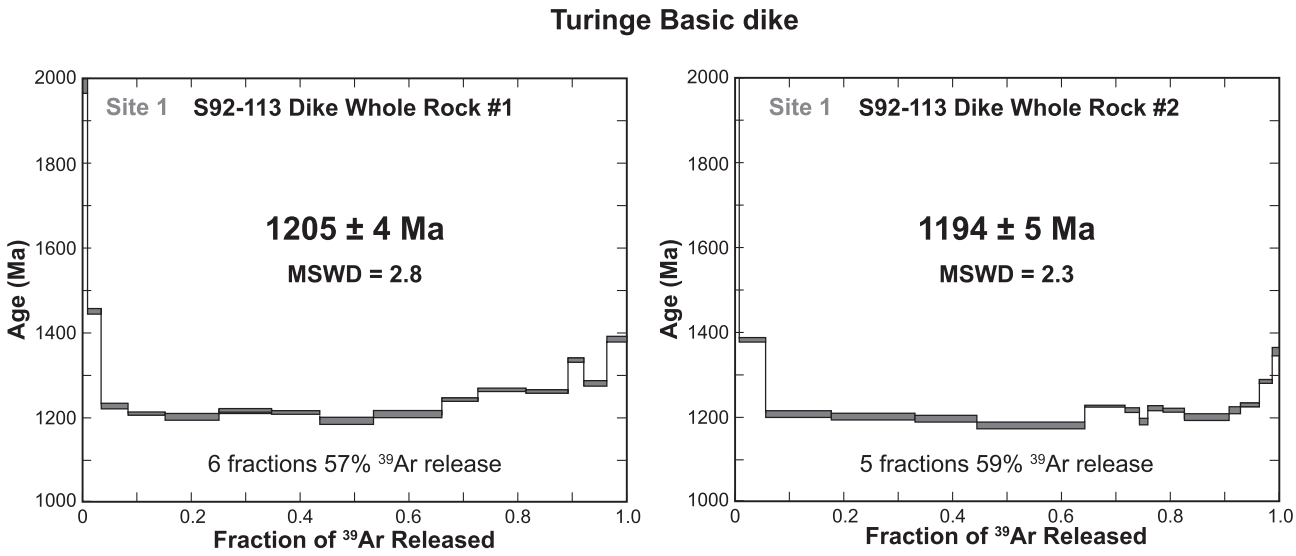


Fig. 9. $^{40}Ar/^{39}Ar$ age spectra from whole rock samples of the Basic dyke cutting the Turinge gabbro-diabase.

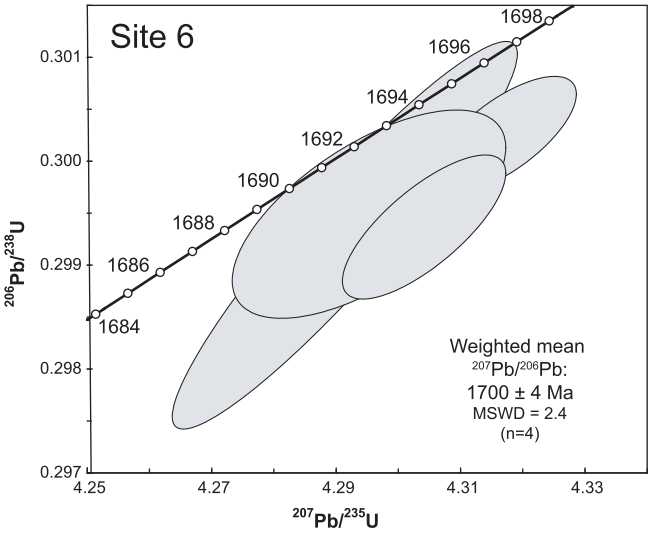


Fig. 10. Concordia plot of U-Pb data from baddeleyite in the Turinge gabbro-diabase. Ellipses indicate error at the 95% confidence level.

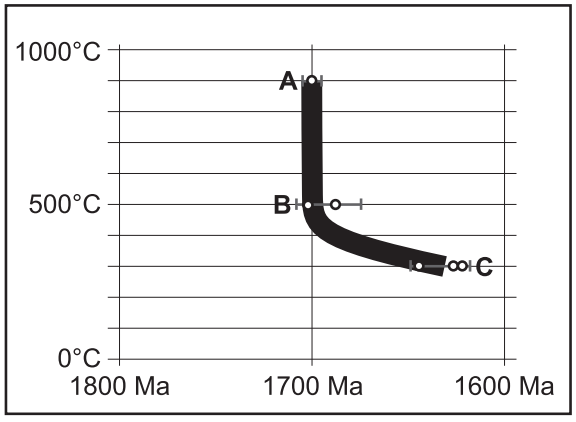


Fig. 11. Suggested cooling behavior of the Turinge gabbro-diabase based on the U-Pb crystallization age and the $^{40}Ar/^{39}Ar$ ages of hornblende and biotite. The closing temperatures of the radiometric system of baddeleyite (A) is ca. 900°, while the closing temperatures of hornblende (B) and biotite (C) are estimated at ca. 500 °C and 300 °C, respectively. The error bars represent the error in age at the 95% confidence level.

(20°N, 102°E) and a rotation of 26° to rotate 1720 Ukrainian pole (29°N, 204°E) to the 1700 Ma pole (52°N, 220°E) of the Turinge gabbro-diabase. This would indicate a somewhat different tectonic pattern when compared with the rotation of of Sarmatia based on the ca 1800–1770 Ma Ukrainian poles. Bogdanova et al. (2006) suggested that post collisional extensional tectonics was of crucial importance from ca 1800 to 1740 Ma and even later at ca. 1710–1670 Ma, which may be reflected by the new paleomagnetic data. The collision and final amalgamation of Sarmatia/Volgo-Uralia and Fennoscandia coincide with the time of assembly of the Nuna supercontinent.

6. Summary and conclusion

The U-Pb age of baddeleyite (1700 ± 4 Ma) and $^{40}Ar/^{39}Ar$ ages of hornblende (1703 ± 6, 1687 ± 13 Ma) and biotite (1622 ± 5; 1645 ± 4; 1627 ± 11 Ma) reveal a cooling history of the gabbro-diabase at Turinge. The cooling is suggested to have taken place in two stages with different cooling rates, one with an estimated rate up to 46–59°/Ma, related to the time of intrusion and another, more slow, ca 2.9 °C/Ma, related to uplift. From this cooling history it can be concluded that the estimated magnetization age of the gabbro-diabase, ca 1690–1700 Ma, is close to that of the crystallization age and the $^{40}Ar/^{39}Ar$ hornblende age. This small time difference between the U-Pb age and age of magnetization is different from the ca 80 Ma that was estimated for the 1786 Ma Hoting gabbro. Based on a supposed fast initial cooling similar to that disclosed from the Turinge gabbro-diabase, the magnetization age of the Hoting gabbro is now suggested to be close to 1780 Ma. This has implications for estimations of magnetization ages also for other deep intrusions.

The new paleomagnetic pole of the Turinge gabbro-diabase fulfills most of the criteria for a reliable pole and is considered a key pole for Fennoscandia. This pole prolong the time span of overlapping poles for Fennoscandia/Baltica indicating only small movements of the shield between ca 1870 and 1700 Ma. The Joulovaara pole is not very reliable, but plots close to other ca 1800 Ma poles of Fennoscandia. The Basic dyke that cuts the Turinge gabbro-diabase was dated at ca 1200 Ma (whole rock, $^{40}Ar/^{39}Ar$) and the virtual geomagnetic pole calculated from its primary magnetization falls into the expected trend of APWP for Baltica.

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Table 4
Selected paleomagnetic poles from the Fennoscandian shield. Q denotes the quality index of Van der Voo (1990).

Code	*Fennoscandian Poles* Formation	Plat (°N)	Plon (°E)	A95 (°)	Age (Ma)	Q	Ref.
1	Vittangi	42.6	227.9	4.9	ca 1890 not direct age	3	Elming (1985), Skiöld (1988)
2	Keuruu dikes	46.0	231.0	5.5	1870 ± ; 1867 ± 8	7	Klein et al. (2016)
3	Notträsk	43.0	220.0	2.8	1806 ± 6	5	Elming (1985), age Sadeghi and Hellström (2017)
4	Kallax	49.0	209.0	3.9	1802 ± 3; 1796 ± 2 (rim)	5	Elming (1994), age of The Ale granite: Öhlander and Schöberg (1991)
5	Joulovaara	39.1	239.0	12.5	ca 1800	2	*this study*; age Hellström and Bergman, unpubl.
6	Småland intrusions	45.7	182.7	8.0	1784–1769	7	Pisarevsky and Bylund (2010)
7	Shoksha Formation	39.7	221.1	4.0	1780–1760	7	Pisarevsky and Sokolov (2001)
8	Ropruchey Sill	39.1	216.6	6.7	1754–1748	5	Fedotova et al. (1999) Lubina et al. (2012)
9	Nabbrejarka gabbro-diabase	50.0	224.0	1.4	< 1810–1780	2	Elming (1994)
10	Dundret/Vassaravaara gabbro	22.6	186.6	6.1	1883 ± 5; 1798 ± 4	2	Elming (1994) (recalculated)
11	Hoting gabbro	43.0	233.3	10.9	1786 ± 10	6	Elming et al. (2009)
12	Turinge gabbro-diabase	51.6	220.2	4.8	1703 ± 6; 1687 ± 13; 1700 ± 4	6	*This study*
13	Sipoo Quartz porphyry	26.4	180.0	9.4	1643–1623	4	Mertanen and Pesonen (1995)
14	Wiborg Quartz porphyry dykes	30.2	175.4	9.4	1641–1621	3	Neuvonen (1986)
15	Åland basic dyke	23.7	191.4	2.8	1576 ± 3	7	Salminen et al. (2016)
16	Post-Jotnian Intrusions	1.8	159.1	3.4	1270–1246	6	Elming and Pesonen (2010)
17	Turinge Basic dyke	3.4	181.6	3.3	ca 1200	4	*This study*
18	Laanila Dolerite	2.1	212.2	13.8	1095–995	3	Mertanen et al. (1996)

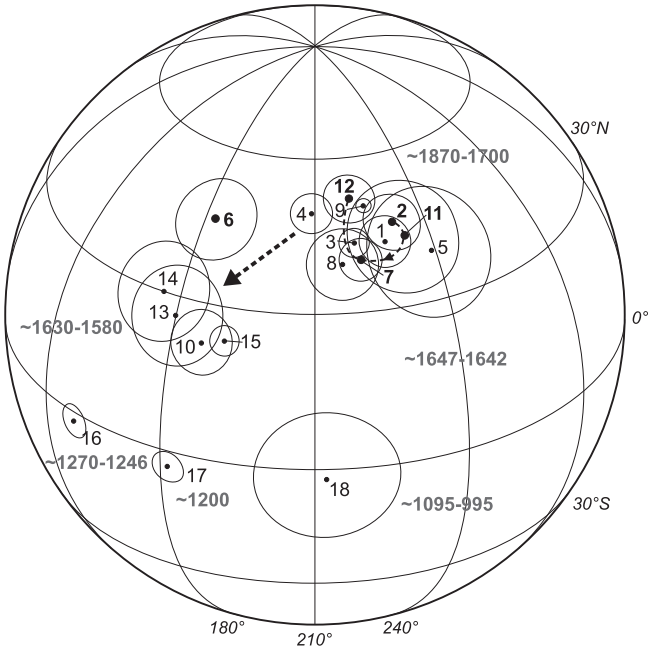


Fig. 12. Dated paleomagnetic poles in time spans from ca. 1870 to 995 Ma for Fennoscandia/Baltica. For the codes of the poles see Table 4. The paleomagnetic poles 5 (Joulovaara gabbro) and 12 (Turinge gabbro-diabase) are from this study as so is also the VGP 17. Based on criteria of Van der Voo (1990), the most reliable poles for the time span 1870 Ma to 1700 Ma are 2, 6, 7, 11, 12 (in bold and a larger black symbol) and for the ca 1650 to 1580 Ma time span the 13, 14, 15 and 16 poles. The deviation of pole 6 from the cluster of 1870–1700 poles is discussed in the text. A possible loop in the Apparent Polar Wander Path is defined by the thin hatched line connecting the poles 2, 7, 11 and 12.

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